Arnav Shukla

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Summary

Full-stack and applied-AI engineer building production-grade systems, from row-level-secured healthcare platforms to autonomous, fault-tolerant AI agents and resource-constrained ML pipelines. End-to-end system development: database design, backend APIs, and deployment. Benchmarks performance, latency, memory, and security.

Technical Skills

Languages	Python, Java, C/C++, JavaScript, TypeScript, SQL
Frontend	React, Next.js, Vue.js, HTML5, CSS3, Tailwind CSS, Zod
Backend	FastAPI, Flask, Django, Node.js, REST APIs, WebSockets
Databases	PostgreSQL, MySQL, SQLite, Redis, Supabase, PostGIS, SQLAlchemy, Row-Level Security (RLS)
Cloud / DevOps	AWS (Cloud Architecting), Docker, Docker Compose, Vercel, Render
AI / ML	Gemini, Groq, Ollama, Hugging Face, scikit-learn, Whisper, spaCy, Prompt Engineering, RAG, Model Context Protocol (MCP)
Developer Tools	Git, GitHub, Postman, Playwright, VS Code, Figma

Projects

Rural Healthcare Platform | *Next.js, TypeScript, Supabase, PostgreSQL, PostGIS, Gemini* | Github · Live

- Built a full-stack bilingual (English/Hindi) telehealth platform with Row-Level Security across **11 PostgreSQL tables**; verified SQL-injection and RLS-bypass attempts return **0 rows** in production
- Shipped a Gemini-powered symptom checker and PostGIS `ST_DWithin` geo-search over **6,616 indexed facilities**, averaging **274ms** query latency (P95 558ms) via a GIST spatial index
- Designed 12 rate-limited REST routes with Zod validation and Jitsi teleconsultation; achieved **100% pass rate** on a 10-point automated security audit (HSTS, CSP, SAMEORIGIN) and **197ms average TTFB** on Vercel Edge (Mumbai)

Eco-Loop: Autonomous Building AI | *FastAPI, React, Ollama/Qwen2.5, FastMCP* | Github   Honeywell Hackathon

- Developed an autonomous HVAC control agent using a locally-hosted Qwen2.5 (1.5B) LLM via Ollama, cutting simulated energy consumption by **50%** (0.86 vs. 1.71 kWh) against a rule-based baseline while maintaining ASHRAE 55 comfort (avg PMV 0.00) across a 5-zone office
- Exposed a **9-tool** Model Context Protocol (MCP) server (avg **0.006ms** execution, 100% success rate) so any MCP-compliant AI client can control the simulated building
- Built a fault-tolerant control loop with automatic rule-based fallback, sustaining **100% uptime with zero crashes** across 30 injected disconnect, timeout, and malformed-response failures, streamed over WebSockets to React dashboard

TalentMatch AI | *Python, FastAPI, scikit-learn, Groq API, Docker* | Github · Live

- Architected a stateless ATS resume-scoring pipeline (`parsing` → `ranking` → `explainability`); TF-IDF feature engineering and semantic ranking execute in **under 21ms** by eliminating deep-learning dependencies
- Dockerized the full service to a **240MB image** with ∼**2.9MB** peak per-request memory overhead, running comfortably within Render's **512MB** free-tier limit at ∼37% CPU per request
- Developed a companion web extension (vanilla JS, no framework overhead) with **under 15ms** total startup time; served via a stateless single-worker API sustaining ∼**24.6 requests/sec**

CampusCart: Multi-Vendor Campus Marketplace | *Python, Flask, SQLAlchemy, SQLite* | Github · Live

- Engineered a multi-tenant marketplace across **3 roles** (Student, Vendor, Admin) using Flask Blueprints, with catalog queries executing in **under 1ms** and **0% error rate** across benchmark testing
- Designed a relational schema with strict foreign-key constraints, Bcrypt-hashed authentication, role-based access decorators, and a simulated campus wallet, running on **under 85MB RAM**

Education

Vellore Institute of Technology - Bhopal — May 2027
Bachelor of Technology, Computer Science and Engineering; CGPA: 8.57 — *Bhopal, MP, India*

- Relevant Coursework: DSA, OOP, OS, DBMS, Computer Networks

Higher Secondary (M.P. Board) – 83.4% — May 2023

Secondary (CBSE) – 94.33% — May 2021

Certifications

- **AWS Academy Graduate – Cloud Architecting** (60 hrs) — *Jun 2026*
- **Machine Learning** – NPTEL/IIT Madras | **Marketing Analytics** – NPTEL/IIT Kharagpur — *2025–26*
- **AI Knowledge Challenge**(winner) – IBM Call for Code | **Computer Networking** – Coursera/Google — *2024*

Extracurricular Activities

- Shortlisted for Honeywell Hackathon (Eco-Loop) round 1; pitched HealthSeva AI at Smart India Hackathon 2025 (college round); active participant in VIT Bhopal tech fest